

High-Order Implicit Heat Equation Solver Verification Report for **example5**

Generated report for `testHighOrderHeatEquationImplicit`

May 7, 2026

Abstract

This report updates the verification documentation for `testHighOrderHeatEquationImplicit` using the current **example5** results in `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/testHighOrderHeatEquationImplicit/results/example5`. Its structure follows the report used for `highOrderManufacturedFDAImplicit`, but the model is simpler: only the scalar heat variable T is advanced, and there are no ionic ODE states. The report compares the standard finite-volume operator with LRE high-order reconstructions, backward Euler and Crank–Nicolson time integration, and lumped or consistent high-order mass matrices.

1 Governing Equation

The solver advances the transient scalar diffusion problem

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (\alpha \nabla T) + Q. \quad (1)$$

For **example5**, the residual uses the same FDA-like manufactured source as the explicit heat solver,

$$R_K = L_h(T)_K - Q_K, \quad (2)$$

where Q is evaluated from an exact manufactured ionic-current expression but no ODE variables are solved.

2 Manufactured Example5

The exact field is

$$T^*(\mathbf{x}, t) = A\sqrt{1+t}F(\mathbf{x}). \quad (3)$$

In 2D,

$$F(x, y) = \cos(\pi x) \cos(2\pi y), \quad G(x, y) = 1 + xy^2. \quad (4)$$

The source uses

$$Q^* = AI_{\text{ion}}^*, \quad (5)$$

$$I_{\text{ion}}^* = -\frac{1}{2}(u_1^* + u_3^* - T^*)(u_2^*)^2(T^* - u_3^*) + \beta_T(T^* - u_3^*), \quad (6)$$

$$u_1^* = (1+t)G + T^*, \quad u_2^* = \frac{1}{(1+t)\sqrt{G}}, \quad u_3^* = 0. \quad (7)$$

The auxiliary u_i^* enter only through the manufactured source.

3 Spatial Discretization

For `useHighOrder.T=false`, the solver uses the standard OpenFOAM cell-centred finite-volume Laplacian. For `useHighOrder.T=true`, the diffusion operator is assembled from LRE reconstructions into a high-order stiffness matrix K and mass matrix M . In semi-discrete notation,

$$M \frac{dT}{dt} = KT + MQ, \quad (8)$$

with the sign of Q adjusted for `example5` in the code. The options `p1`, `p2`, and `p3` select first-, second-, and third-order reconstruction terms, while `N0` uses the standard second-order finite-volume operator.

4 Mass Matrices

The high-order solve can use a lumped or consistent mass matrix. The lumped option reduces the mass to diagonal cell-volume weights. The consistent option integrates the reconstruction basis over cell quadrature points, which couples stencil coefficients through the mass matrix. For standard `N0` runs, only the lumped finite-volume mass is used in the current results.

5 Boundary Conditions

For the high-order flux/operator, `zeroGradient` and `empty` patches are treated as no-flux boundaries,

$$\int_{\partial K \cap \Gamma_N} \mathbf{n} \cdot \alpha \nabla T \, dS = 0. \quad (9)$$

Fixed-value patches keep their boundary contribution in the high-order matrix/source assembly.

6 Implicit Time Integration

The solver uses a theta method,

$$\theta = 1 \quad \text{for backward Euler}, \quad \theta = \frac{1}{2} \quad \text{for Crank-Nicolson}. \quad (10)$$

A compact form of the linear heat solve is

$$\left(\frac{M}{\Delta t} - \theta K \right) T^{n+1} = \left(\frac{M}{\Delta t} + (1 - \theta)K \right) T^n + \theta M Q^{n+1} + (1 - \theta) M Q^n, \quad (11)$$

with the appropriate source sign for `example5`.

7 Error Measures and Rates

The solver writes cell-centred errors L_1 , L_2 , L_∞ and reconstructed quadrature errors G_1 , G_2 , G_∞ . The tables below recompute the L_2 and G_2 log-log rates directly from the `errors_vs_N` files. The expected spatial orders are approximately 2 for `N0` and `p1`, 3 for `p2`, and 4 for `p3`.

8 2D Results

The available 2D data are read from `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/testHighOrderHeatEquationImplicit/results/example5/2D`. Tables report fitted orders over $N = 10$ – 100 and $N = 70$ – 100 .

Table 1: Spatial convergence orders for the implicit heat solver, 2D example5, $\Delta t = 10^{-3}$. Fits use $N = 10$ –100 and $N = 70$ –100. Schemes: bE = backward Euler, cN = Crank–Nicolson. Mass: cons. = consistent, lump. = lumped.

HO T	Scheme	Mass	Exp.	$p_{L_2}^{10-100}$	$p_{G_2}^{10-100}$	$p_{L_2}^{70-100} / p_{G_2}^{70-100}$
N0	bE	lump.	2	2.037	2.037	2.134 / 2.134
N0	cN	lump.	2	1.997	1.997	2.000 / 2.000
p1	bE	cons.	2	2.410	2.342	2.474 / 2.336
p1	bE	lump.	2	2.410	2.342	2.474 / 2.336
p1	cN	cons.	2	2.406	2.338	2.444 / 2.311
p1	cN	lump.	2	2.406	2.338	2.444 / 2.311
p2	bE	cons.	3	2.131	2.208	2.065 / 2.079
p2	bE	lump.	3	2.127	2.207	2.066 / 2.081
p2	cN	cons.	3	2.120	2.196	2.025 / 2.039
p2	cN	lump.	3	2.115	2.195	2.025 / 2.039
p3	bE	cons.	4	3.212	3.511	0.098 / 0.174
p3	bE	lump.	4	2.661	2.973	1.056 / 1.005
p3	cN	cons.	4	4.084	4.118	4.476 / 4.369
p3	cN	lump.	4	3.062	3.381	2.015 / 2.018

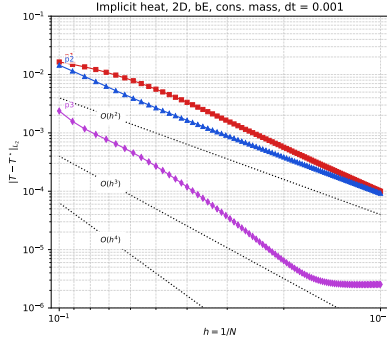
Table 2: Spatial convergence orders for the implicit heat solver, 2D example5, $\Delta t = 10^{-4}$. Fits use $N = 10$ –100 and $N = 70$ –100. Schemes: bE = backward Euler, cN = Crank–Nicolson. Mass: cons. = consistent, lump. = lumped.

HO T	Scheme	Mass	Exp.	$p_{L_2}^{10-100}$	$p_{G_2}^{10-100}$	$p_{L_2}^{70-100} / p_{G_2}^{70-100}$
N0	bE	lump.	2	2.001	2.001	2.012 / 2.012
N0	cN	lump.	2	1.997	1.997	2.000 / 2.000
p1	bE	cons.	2	2.406	2.338	2.447 / 2.313
p1	bE	lump.	2	2.406	2.338	2.447 / 2.313
p1	cN	cons.	2	2.406	2.338	2.444 / 2.311
p1	cN	lump.	2	2.406	2.338	2.444 / 2.311
p2	bE	cons.	3	2.121	2.197	2.029 / 2.043
p2	bE	lump.	3	2.116	2.196	2.029 / 2.043
p2	cN	cons.	3	2.120	2.196	2.025 / 2.039
p2	cN	lump.	3	2.115	2.195	2.025 / 2.039
p3	bE	cons.	4	4.076	4.162	4.084 / 4.523
p3	bE	lump.	4	3.009	3.327	1.848 / 1.838
p3	cN	cons.	4	4.084	4.118	4.478 / 4.370
p3	cN	lump.	4	3.062	3.381	2.014 / 2.017

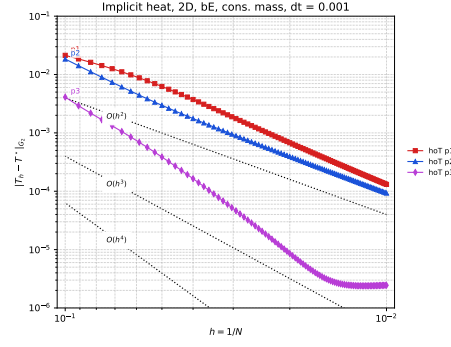
8.1 2D Convergence Figures

$$\Delta t = 10^{-3}$$

$$\Delta t = 10^{-4}$$

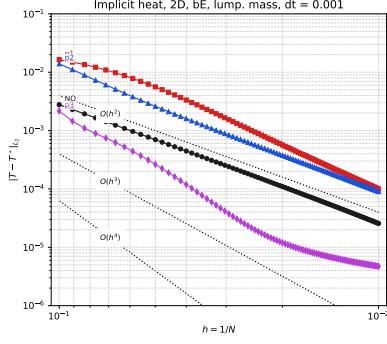


(a) $\|T - T^*\|_{L_2}$

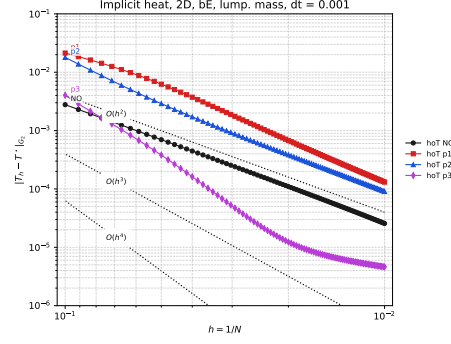


(b) $\|T_h - T^*\|_{G_2}$

Figure 1: Implicit high-order heat-equation solver, 2D, bE with cons. mass, $\Delta t = 10^{-3}$. The left panel shows the cell-centred L_2 error and the right panel shows the reconstructed quadrature G_2 error.

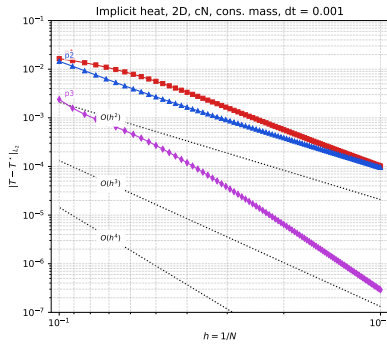


(a) $\|T - T^*\|_{L_2}$

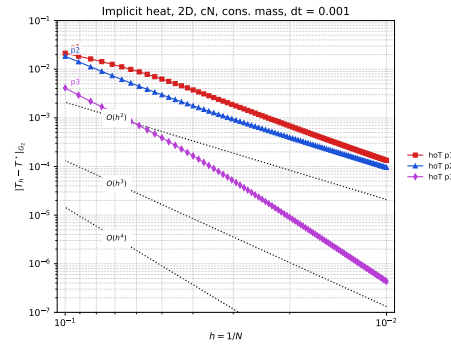


(b) $\|T_h - T^*\|_{G_2}$

Figure 2: Implicit high-order heat-equation solver, 2D, bE with lump. mass, $\Delta t = 10^{-3}$. The left panel shows the cell-centred L_2 error and the right panel shows the reconstructed quadrature G_2 error.

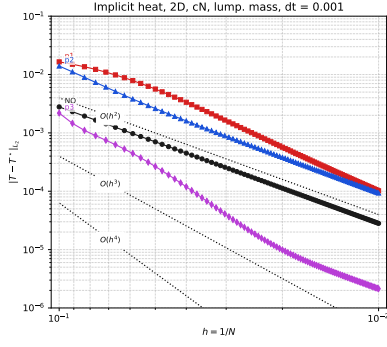


(a) $\|T - T^*\|_{L_2}$

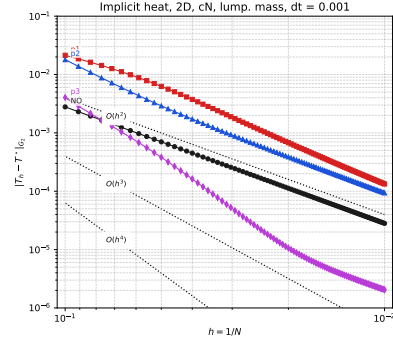


(b) $\|T_h - T^*\|_{G_2}$

Figure 3: Implicit high-order heat-equation solver, 2D, cN with cons. mass, $\Delta t = 10^{-3}$. The left panel shows the cell-centred L_2 error and the right panel shows the reconstructed quadrature G_2 error.

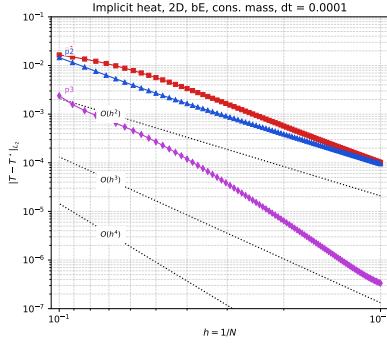


(a) $\|T - T^*\|_{L_2}$

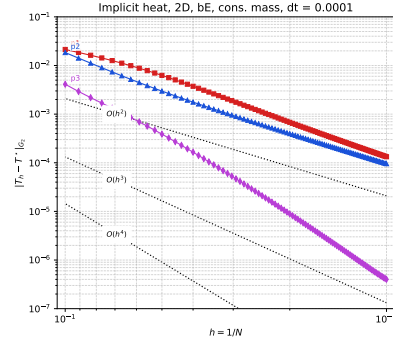


(b) $\|T_h - T^*\|_{G_2}$

Figure 4: Implicit high-order heat-equation solver, 2D, cN with lump. mass, $\Delta t = 10^{-3}$. The left panel shows the cell-centred L_2 error and the right panel shows the reconstructed quadrature G_2 error.

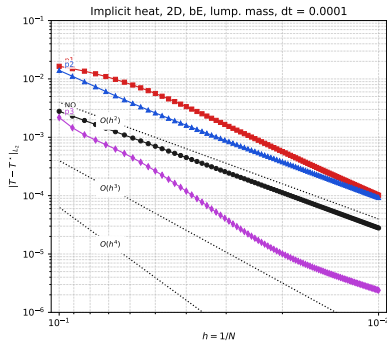


(a) $\|T - T^*\|_{L_2}$

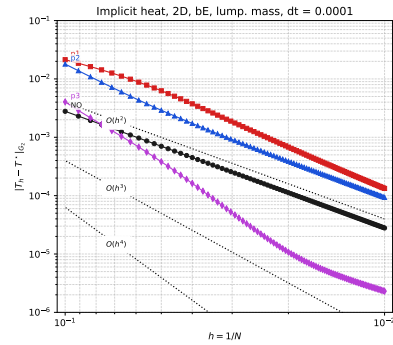


(b) $\|T_h - T^*\|_{G_2}$

Figure 5: Implicit high-order heat-equation solver, 2D, bE with cons. mass, $\Delta t = 10^{-4}$. The left panel shows the cell-centred L_2 error and the right panel shows the reconstructed quadrature G_2 error.

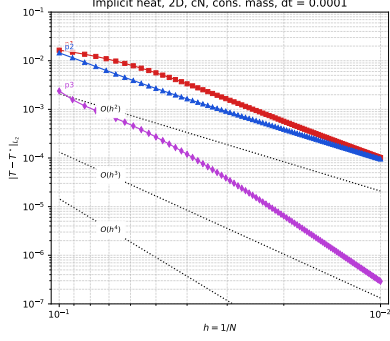


(a) $\|T - T^*\|_{L_2}$

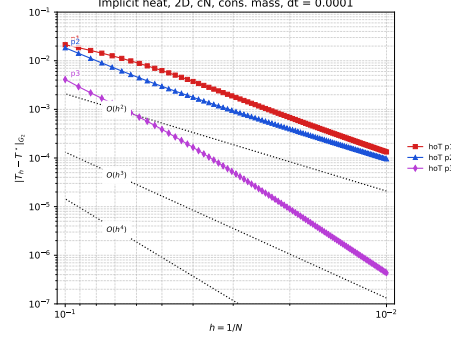


(b) $\|T_h - T^*\|_{G_2}$

Figure 6: Implicit high-order heat-equation solver, 2D, bE with lump. mass, $\Delta t = 10^{-4}$. The left panel shows the cell-centred L_2 error and the right panel shows the reconstructed quadrature G_2 error.

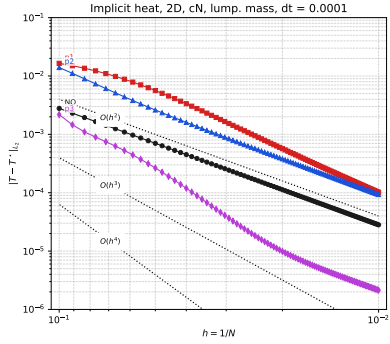


(a) $\|T - T^*\|_{L_2}$

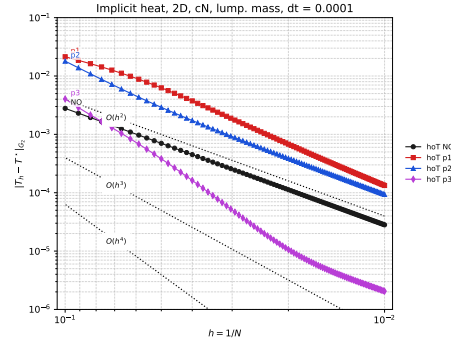


(b) $\|T_h - T^*\|_{G_2}$

Figure 7: Implicit high-order heat-equation solver, 2D, cN with cons. mass, $\Delta t = 10^{-4}$. The left panel shows the cell-centred L_2 error and the right panel shows the reconstructed quadrature G_2 error.



(a) $\|T - T^*\|_{L_2}$



(b) $\|T_h - T^*\|_{G_2}$

Figure 8: Implicit high-order heat-equation solver, 2D, cN with lump. mass, $\Delta t = 10^{-4}$. The left panel shows the cell-centred L_2 error and the right panel shows the reconstructed quadrature G_2 error.

9 Three-Dimensional Results

No `example5/3D` result directory was present when this report was generated. The report and plotting script will pick up three-dimensional data if they are added later with the same folder layout.

10 Discussion

The current `example5` result set contains two-dimensional runs. The report keeps the implicit FDA report structure but collapses the variables to the single heat unknown T . The tables make it possible to separate spatial behaviour from temporal/mass-matrix effects by comparing backward Euler versus Crank–Nicolson and lumped versus consistent mass at fixed Δt .

11 Files

- Solver source: `/home/pablo/cardiacFoam-pc/applications/solvers/testHighOrderHeatEquationImplicit.C`
- Tutorial results: `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/testHighOrderHeatEquationImplicit/results/example5`
- Plot script: `/home/pablo/cardiacFoam-pc/applications/solvers/testHighOrderHeatEquationImplicit/report/plot_convergence.m`